

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/(CE/EE/ME/ECE) / BS-M201B/ 2024-25
2025
MATHEMATICS-IIB

24101101063.

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

- 5×2=10
1. Using Green's theorem, show that $\frac{1}{2} \oint_C (x dy - y dx) = \text{area of the region enclosed by the closed curve } C$. 2
 2. Show that the functions $1, x, x^2$ are linearly independent. 2
 3. Find an integrating factor of $(xy^2 - e^{\frac{1}{x^3}}) dx - x^2 y dy = 0$ 2
 4. Show that the function $f(z) = |z|^2$ is not analytic at the point $z = 0$. 2
 5. Show that $\int_{\Gamma} dz = \beta - \alpha$ where Γ is any rectifiable curve joining α and β . 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *five* questions

- 12×5 = 60
6. (i) Evaluate $\iint_R \sqrt{4a^2 - x^2 - y^2} dx dy$ where R is the upper half of the circle $x^2 + y^2 = 2ax = 0$. 5
 - (ii) Evaluate $\int_C (\cos x \sin y - xy) dx + \sin x \cos y dy$ by Green's theorem in the plane, where C is the circle $x^2 + y^2 = 1$ in the xy -plane described in the positive sense. 5
 - (iii) Change the order of integration: $\int_0^1 dy \int_y^1 e^{x^2} dx$. 2
 7. (i) Solve: $y(xy + 2x^2 y^2) dx + x(xy - x^2 y^2) dy = 0$. 4
 - (ii) Solve: $x \frac{dy}{dx} + y = y^2 \log x$. 4
 - (iii) Find the general solution and the singular solution of the equation $y = px + \sqrt{9p^2 + 25}$ where $p = \frac{dy}{dx}$. 2+2
 8. (i) Evaluate $\iint_S \vec{F} \cdot d\vec{s}$ where $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ and S is the part of the surface of the sphere $x^2 + y^2 + z^2 = 1$ which lies in the first quadrant. 6
 - (ii) State Stoke's theorem. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ by Stoke's theorem, where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices at $(0,0,0)$, $(1,0,0)$, $(1,1,1)$. 6
 9. (i) Solve: $(D^2 - 2D)y = e^x \sin x$. 4
 - (ii) Solve: $\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} - 4y = xe^{-2x}$ 4

$2D+1=0$
 $2D=-1$
 $D=-1/2$

10. (iii) Solve: $(D^3 - 1)y = x \sin x$ 4
- (i) Solve by the method of variation of parameters: $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x$ 6
11. (ii) Solve: $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \log x$ 6
- (i) Examine whether the function $f(z)$ defined by $f(z) = \begin{cases} \frac{xy(y-ix)}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ satisfies Cauchy-Riemann equations at the origin or not. Also test the differentiability of the function at the origin. 2+2
- (ii) If $f(z)$ is an analytic function of z , prove that $\nabla^2 [Re f(z)]^2 = 2 |f'(z)|^2$ where $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$. 4
- (iii) Show that the function $u(x, y) = y^3 - 3x^2 y$ is harmonic and find its harmonic conjugate $v(x, y)$ such that $f(z) = u + iv$ is analytic. 2+2
12. (i) Find the bilinear transformation which maps the points $z = 1, 0, -1$ into the points $w = i, 0, -i$ respectively and find the fixed points of the transformation. 3+1
- (ii) Expand the function $f(z) = \log(1+z)$ in Taylor's series about $z = 0$ and indicate its region of convergence. 3+1
- (iii) Expand the function $f(z) = \frac{-1}{z^2 - 3z + 2}$ as a Laurent's series in the annular region $1 < |z| < 2$. 4
13. (i) Use Cauchy's residue theorem to evaluate $\int_{\Gamma} \frac{e^z}{z^2 + \pi^2} dz$ where Γ is the positively oriented circle $|z| = 4$. 4
- (ii) Use Cauchy's integral formula to evaluate $\int_{\Gamma} \frac{e^z}{z^2 + 4} dz$ where Γ is the positively oriented circle $|z - i| = 2$. 4
- (iii) By Contour integration, evaluate $\int_0^{2\pi} \frac{1}{2 + \cos \theta} d\theta$ 4

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[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CE/EE/ME/ BS-CH201/ 2024-25
2025
CHEMISTRY

40

Full Marks: 70

Times: 3 Hours

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Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5x2=10

1. Define crystal field stabilization energy (CFSE). 2
2. What do you mean by the term 'auxochrome' in spectroscopy? Give example. 2
3. At 673 K temperature carbon monoxide is an effective reductant than Carbon-justify the comment. 2
4. What is tautomerism? 2
5. Give example of an optically active compound without chiral carbon. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions

4x15=60

6. i) Derive Schrodinger wave equation using de Broglie equation. 3
 ii) What is the concept of particle in a one-dimensional box? What is Schrodinger wave equation in such a case? How can this equation be solved for ψ and E ? 2+2+3
 iii) What will happen if the walls of the one-dimensional box are suddenly removed? 2
 iv) Calculate the probability of finding the particle between $0.49a$ and $0.5a$ for the states ψ_1 and ψ_2 ? 3
7. i) Define the term screening constant? Calculate the effective nuclear charge for the electron of penultimate shell of an atom whose atomic number is 38. 1+3
 ii) What are CFSE for the following system: d^1 octahedral, d^5 low spin and high spin octahedral, d^8 low spin and high spin octahedral. 3
 iii) In what respect does an inner orbital complex differ from an outer orbital complex? Colour of the coordination complexes can be explained by the crystal field theory – comment on the statement. 2+3
 iv) How does the boiling point of molecules affected by van der Waals dispersion forces? Explain with examples. 3
8. i) Derive the thermodynamic relation $(\Delta G)_T = nRT \ln V_2/V_1$ for an ideal gas. (the thermodynamics functions have their own significance). 2
 ii) Prove that in an irreversible process, net entropy change for the system and surroundings is greater than zero. 2
 iii) The emf of the cell : $\text{Cd} / \text{CdCl}_2 (1\text{M}) // \text{AgCl} / \text{Ag}$, is 0.675 volt at 25°C . The temperature coefficient of the cell is -6.5×10^{-4} volt/degree. Find ΔH and ΔS 3
 iv) Selection of a reductant in metallurgy is a subject of thermodynamic factor. How do you support this statement. Give two examples with Ellingham diagram to support of your answer? 4
 v) What are conditions for a reaction to be spontaneous in terms of ΔG , ΔH and ΔS ? The values of ΔG° for the formation of Cr_2O_3 and Al_2O_3 are -540 kJ/mol , -827 kJ/mol respectively. So, reduction of Cr_2O_3 by aluminum metal is possible? Explain 2+2
9. i) Explain the stability order of the ions O_2^+ , O_2^- , O_2^{2-} by the help of molecular orbital diagrams. 3
 ii) The Be and N in second period and Mg and P in the third period of the table have slightly higher ionization energies than expected. Explain. 2
 iii) Compare the solubility of CuCl and LiCl in water. 2

iv) Write down the basic principles of soft-hard acid base theory. Explain the following reaction based on 2+2

SHAB theory: $\text{CsF} + \text{LiI} \rightarrow \text{CsI} + \text{LiF}$

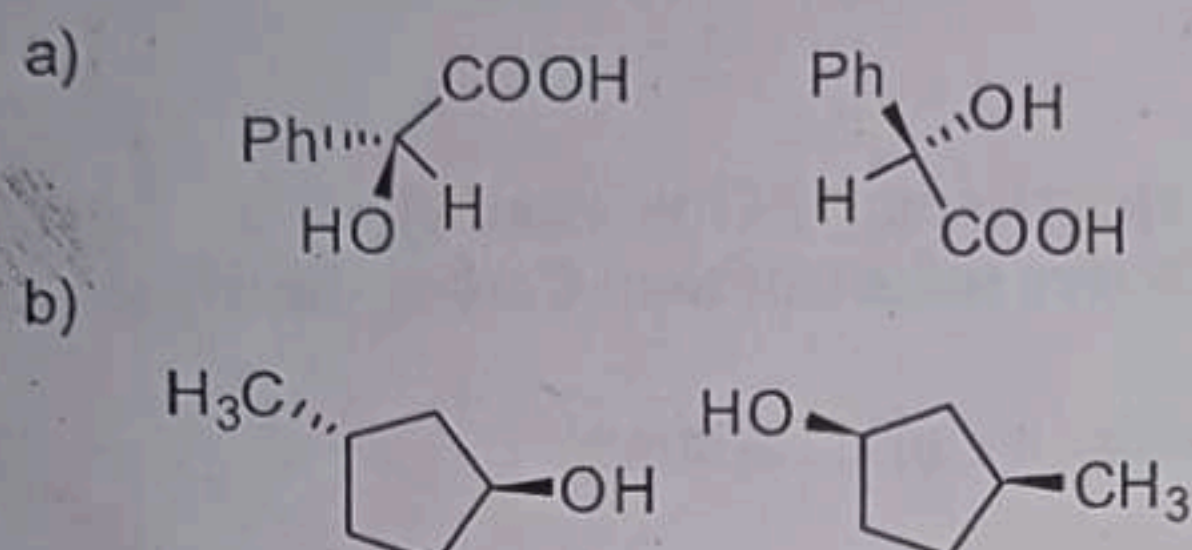
Define electronegativity. Calculate the electronegativity of C in C-H bond if $E_{\text{C-H}}$, $E_{\text{H-H}}$, and $E_{\text{C-C}}$ are 98.8, 104 and 83 kJ/mol respectively. 1+3

10. i) Explain the conformational analysis of butane with energy diagrams. 4
 ii) Explain the following order of stability in carbanions:

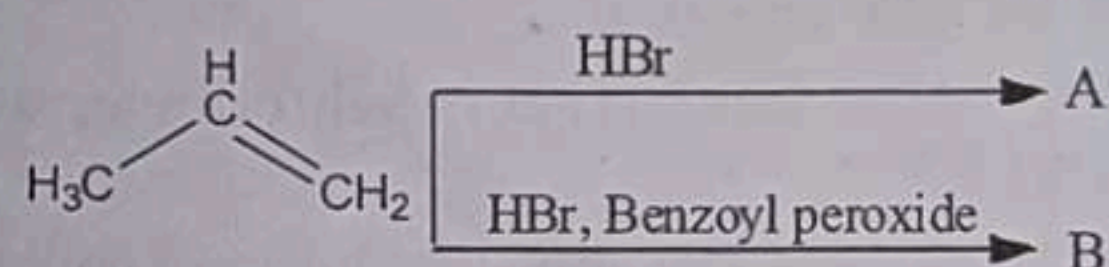


iii) Differentiate between $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions based on mechanism and stereochemistry. 4

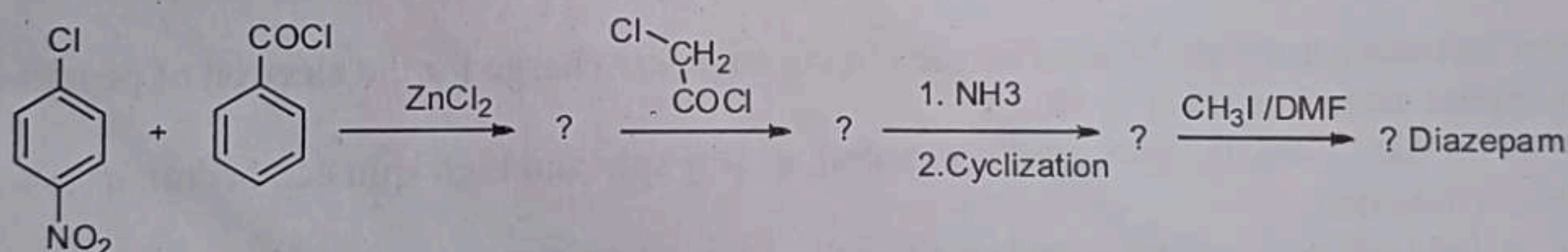
iv) What are the stereochemical relationships between the pair of compounds? Assign R/S configurations at the stereogenic center. 2+2



11. i) Identify the product (A and B) of the following reactions with proper mechanism: 4



- ii) Complete the following reactions and write the name of reagents where require for the synthesis of diazepam 4



iii) Give one example of each: (a) Ionisation isomer (b) Linkage isomer 2

iv) A solution of a chiral compound was measured with a polarimeter. The observed rotation was 5° for a path length of 10 cm. The specific rotation of the compound is known to be 20° . Calculate the concentration of the solution in g/mL. 3

v) Draw the Fischer projection formula of (2S,3R)-3-aminobutan-2-ol. 2

12. Write short notes on **any three** of the following: 3x5

(i) Fajan's rule (ii) Gibbs Helmholtz equation (iii) Hydrogen bonding (iv) Mesomeric effect and Hyperconjugation effect (v) Carbocation and Carbanion

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JGEC/B.TECH/ CE/EE/ME/CSE/ECE/IT/ HM-HU201/ 2024-25

ENGLISH

Full Marks: 70

Time: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer **all** questions

Correct the error in the sentences.

5x2=10

1. The captain along with his team are practising very hard for the forthcoming matches. 2
2. Supposing if it rains what shall we do? 2
3. Everyone knows that the leopard is the faster of all animals. 2
4. Ram was senior than Sam in college. 2
5. It was him who came running into the classroom. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any **four** questions

4x15 = 60

6. i) Communication is a process of sending and receiving information. Explain the communication process in the light of this statement. 10
- ii) Fill in the blanks with appropriate prepositions 5
 - a) When you believe injustice is being done stand _____ the authority.
 - b) She puts _____ two hours on her English studies every day.
 - c) Your strategy ultimately paid _____ and we are reaping the benefits.
 - d) She excels _____ mathematics.
 - e) Please shut the door _____ you.
7. i) Wanted a Plant Manager (operations) at our new factory in Gaziabad, UP. Engineering Graduates with minimum 5 years' experience in manufacturing industries can apply. Salary negotiable. Apply with particulars to Box 650, The Hindu, Chennai-600004. 10
- ii) One-word Substitutions: 5
 - a) A famous person:
 - b) Divide into two parts:
 - c) Extreme fear of being in a small enclosed place:
 - d) One who runs own business:
 - e) Belonging to the same period:
8. i) Realizing the need of packing services in Faridabad, Elite Professional Packers have recently started their services in the city. You, as the publicity manager of this company have the onus of promoting this service. Draft a sales letter to be sent to the prospective users. 10
- ii) Use the correct form of tense in the blanks 5

Marie had probably caught my odd stare at the table and so _____ (move) over and blew the candle out. She then _____ (glance) at the window and _____ (go) over to open them. Father was _____ (sit) on the bed and said in a cheerful voice, "I _____ (like) the country air. It suits me well."
9. i) Imagine yourself to be the instructor of a course in which 75 students have registered. Draft an email to be sent to all these students asking them to select a topic of their choice and prepare for a professional presentation. 10
- ii) Transform the sentences according to the instruction given: 5
 - a) Horses are bigger than donkeys. (begin: Donkeys aren't...)

- b) Few people know me. (begin: Not...)
- c) Children drink more milk than adults. (begin: adults don't...)
- d) There is little water. I can't have a bath. (begin: there isn't enough...)
- e) Reading isn't as interesting as travelling. (begin: travelling is...)

10. i) Write an essay on the future of engineering education. 10
 ii) Make sentences to point out the difference between the following pair of words: 5
 a) which/witch b) vein/vain c) yoke/yolk d) weak/week e) rain/reign

11. i) Write a note on the various levels of communication. 10
 ii) Choose the correct synonym of the underlined words from the list given below: 5
 (brushed, grasped, disagreed, steered, soil)
- a) I was digging in the dirt when my mother called me for lunch. *soil*
 - b) Mike groomed his dog. *brush*
 - c) Noah argued with his brothers. *disagree*
 - d) Matt grabbed her hand. *grasped*
 - e) Rob drove his truck to school. *steered*

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JGEC/B.TECH/EE-CE-ME/ES-EE-201/2024-25
BASIC ELECTRICAL ENGINEERING

Times: 3 Hours

Full Marks: 70

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer *all* questions

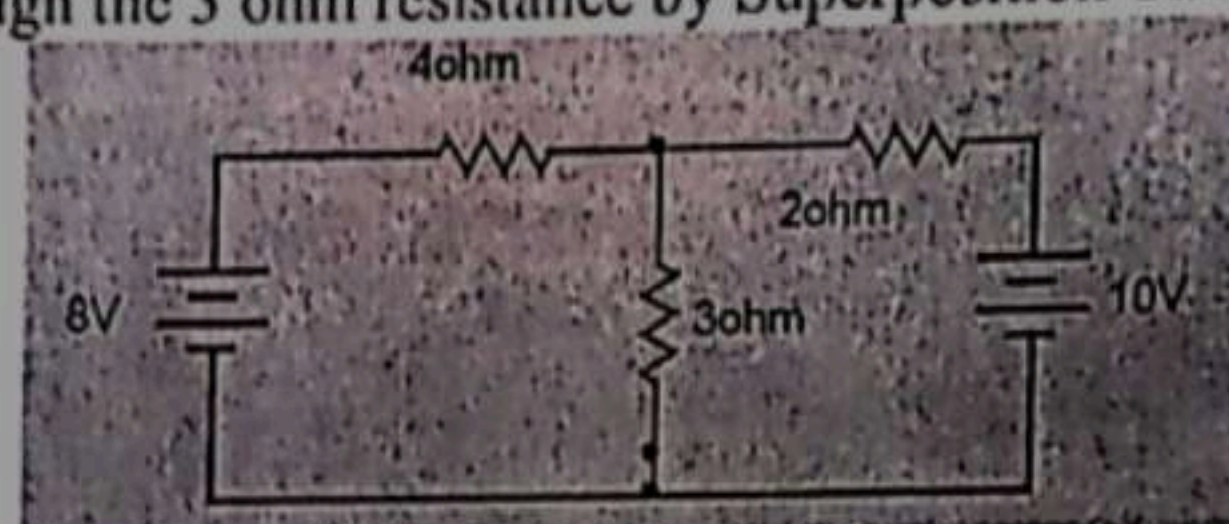
1. Define Linear & Bilateral elements with examples. 2
2. What do you mean by Form-Factor & Peak-Factor in AC circuits? 2
3. Why 3 Phase Induction Motor is called an Asynchronous motor? 2
4. What is the function of Commutators in DC machines? 2
5. "Core loss is a constant loss whereas Coppers loss is a variable loss in single phase transformers"- Justify. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15=60

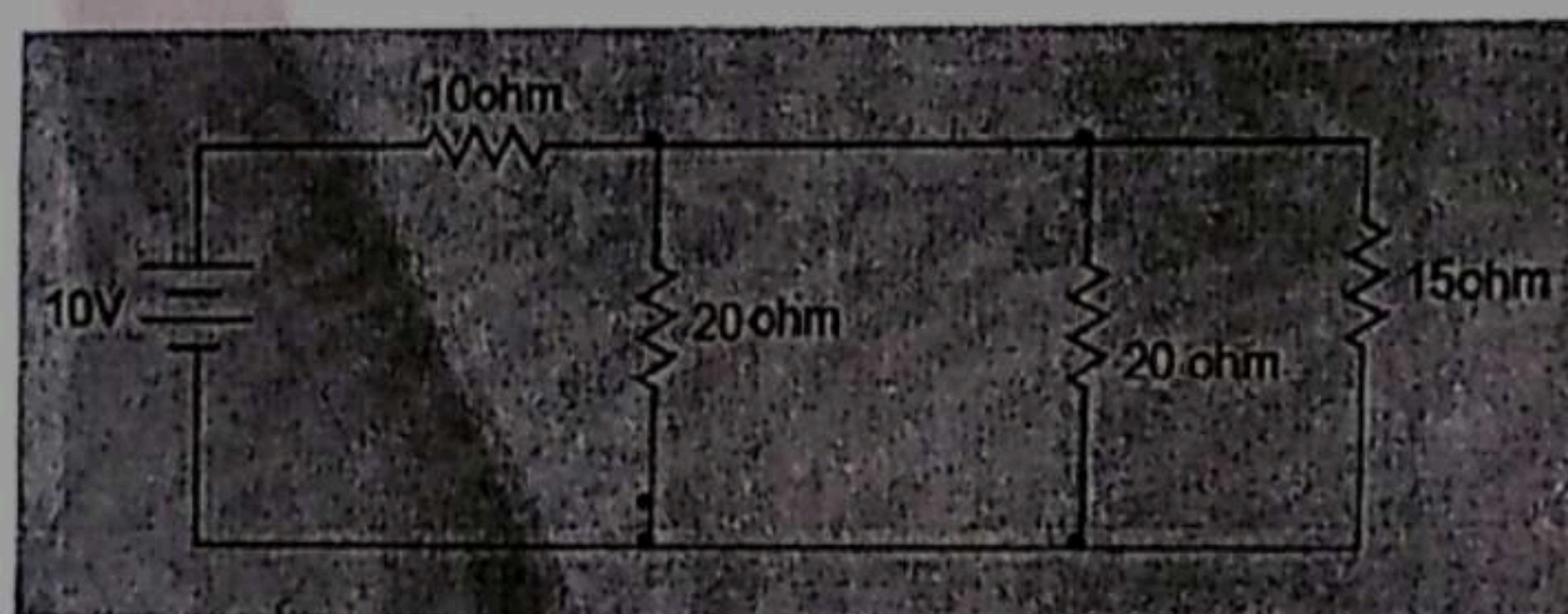
Answer any *four* questions

6. a) Determine the current through the 3 ohm resistance by Superposition Theorem & verify.



6

- b) State Thevenin's Theorem. Determine the current through the 15-ohm resistor in the network given, by Norton's Theorem.



2+7

7. a) Explain i) active & reactive power ii) power factor ii) apparent power for a single phase AC circuit: 5
- b) The equation of an alternating current is $i = 62.35 \sin(323 t)$ A. Determine its
- i) Maximum value
 - ii) Frequency
 - iii) R.M.S. value 5
 - iv) Average value, and
 - v) Form factor
- c) A coil takes a current of 2A when connected to a 240 V, 50 Hz sinusoidal supply and consumes 200 W.
- Calculate the resistance, impedance and inductance of the coil. 5
-
8. a) Explain the principle of operation of a single-phase transformer under loaded condition. 5
- b) The primary winding of a single-phase transformer connected to a 500V(rated voltage), 50Hz supply takes 1.41A and absorbs 125W with the secondary winding open circuited. The secondary open circuit voltage is 250V. When the secondary winding is short-circuited and the primary is connected to a 250V, 5

- 50Hz supply, the primary current is 15.1A (rated current) and the power absorbed is 92W. Determine the shunt and series components of the equivalent circuit.
- c) A 11kV/230 V, 150-kVA single-phase transformer has a core loss of 1.4 kW and a full load copper loss of 1.6 kW. Determine (i) the kVA load for maximum efficiency and (ii) the maximum efficiency at unity power factor load. 5
9. a) A 3-phase 6 pole 50 Hz induction motor has a slip of 1 % at no load and 3 % at full load. Calculate i) Synchronous speed (ii) no load speed (iii) full load speed iv) frequency of rotor current at standstill (v) frequency of rotor current at full load. 5
- b) Find the condition to obtain maximum torque in three-phase induction motor; also find the expression for maximum torque. 5
- c) "A three phase induction motor is self starting"- Explain. 5
10. a) Draw and explain the speed-torque characteristics of a separately excited d.c. motor. 5
- b) Derive the EMF equation of a DC machine. 5
- c) The armature of a 6 pole DC machine has 125 turns and runs at 1000 rpm. The EMF generated on open circuit is 500 Volts. Find the useful flux per pole when the armature winding is (a) Lap connected and (b) wave connected. 5
11. Write short notes on the followings (*any three*):
- a) MCB
 - b) DC-DC Converter
 - c) Core & shell type of Transformers
 - d) Faraday's Law of Electromagnetic Induction.
 - e) No load Test of single-phase transformers.
- 3x5